OpenDiscoveryTrace: Process Traces for Evaluating AI Scientist Workflows

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Abstract

We propose OPENDISCOVERYTRACE, a dataset of 522 AI scientific agent trajectories capturing *how* models reason—not just what they produce. Each trajectory records 9 fields per step across 124 tasks in four science domains, executed by 7 models (3 frontier, 4 open-weight). Pilot analysis shows identical success rates ($\sim$69%) yet 30$\times$ error divergence between models—a dimension invisible to output-only benchmarks. Released under CC-BY-4.0.

1. Scientific Bottleneck

AI systems now conduct scientific research at scale (Jumper et al., 2021; Merchant et al., 2023; Szymanski et al., 2023), and LLM-based agents orchestrate full research pipelines (Lu et al., 2024; Boiko et al., 2023). Benchmarks have emerged to evaluate these systems (Chen et al., 2025; Majumder et al., 2024; Starace et al., 2025; Huang et al., 2024), yet every one evaluates only *final outputs*—generated code, hypotheses, or rubric scores. The reasoning process is discarded.

This creates three problems. First, evaluating only outputs conflates reasoning quality with outcome luck. Second, governance requires complete decision traces for auditing (Kapoor & Narayanan, 2023; Hung & Yen, 2024; Tom et al., 2024). Third, improving agents requires understanding failure modes, which are trajectory properties, not output properties. OPENDISCOVERYTRACE addresses this gap with the first public dataset of structured AI scientific process traces.

2. Proposed Dataset

OPENDISCOVERYTRACE comprises **522 trajectories across 7 models**: 372 from three frontier models (GPT-5.4, Claude Opus 4.6, Gemini 3.1 Pro; 124 each, balanced across 4 domains $\times$ 3 difficulties), 120 from four open-weight models (Qwen2.5-7B, Mistral-7B-v0.3, Phi-3.5-mini, Qwen2.5-1.5B; 30 each), and 30 live-retrieval variants.

Each step records 9 fields extending ReAct (Yao et al., 2023): `phase`, `thought`, `action` (tool/input/output), `observation`, `error`, `revision_trigger`, `confidence`, `step_id`, `timestamp`. Five benchmark tasks are defined: outcome prediction, error localization, claim verification, autonomy classification, and process quality scoring. Full schema in Appendix A.

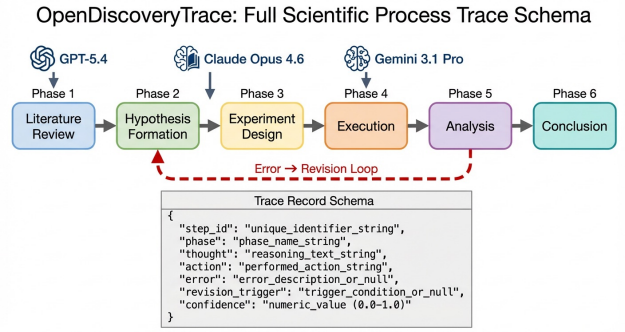

Figure 1. OPENDISCOVERYTRACE: 7 models execute tasks through a six-phase workflow. Each step is a 9-field trace. The 522 trajectories form a corpus of AI scientific process traces.

Why process traces matter. All three frontier models achieve indistinguishable success rates ($\sim$69%, p=0.997), yet Claude Opus 4.6 produces 30$\times$ more errors than GPT-5.4 (2.5 vs. 0.08; H=122, $p<10^{-4}$; Cliff's δ=0.613). These errors are qualitatively different: Claude's are 66.7% tool misuse; GPT-5.4's are 83.6% reasoning errors. This finding is robust across all four domains in matched-task comparisons (Appendix F). LSTM and Transformer sequence-model baselines on outcome prediction match the 69.3% majority baseline (AUROC$\leq$0.42), confirming the task is non-trivial (Appendix H). Output-only benchmarks miss process-level differences entirely.

3. Acquisition Roadmap

Phase 1 (current). 522 trajectories across 7 models (3 frontier + 4 open-weight), 124 tasks, 4 domains, plus 30 live-retrieval variants. Total cost: $\sim$\$340 (API + GPU). All

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infrastructure—schema, harness, analysis pipeline, and five benchmark tasks—is operational and tested.

Phase 2 (months 1–2). Complete all 200 designed tasks across all 7 existing models plus 2 additional open-weight models (Llama 3.1-8B, Gemma-2-9B). Target: 1,800+ trajectories across 9 models, enabling robust cross-architecture comparison at scale with sufficient statistical power for domain-stratified and difficulty-stratified analyses.

Phase 3 (months 2–3). Human expert annotation on a 120-trajectory stratified sample. Three domain-expert annotators independently score five axes: correctness, reasoning quality, tool use efficiency, autonomy level (L1–L4), and error step localization. Target: Krippendorff's $\alpha \geq 0.67$. An LLM-based annotation pilot already achieves α=0.63–0.82 across axes (Appendix I), suggesting human agreement is attainable.

Phase 4 (months 3–6). Live retrieval integration for all literature-domain tasks via PubMed and PubChem APIs (a 30-task pilot is already included). ELN integration pilot for physical-lab trace capture (Dirnagl & Bhatt, 2016; Abolhasani & Kumacheva, 2023), extending the dataset from computational-only to hybrid computational-physical settings.

Sustainability. Semver-versioned schema (v1.0 = current release). Dataset hosted on a public repository with DOI for citability. Open-source harness enabling community-contributed trajectories. Annual refresh cycles incorporating new frontier and open-weight models as they become available.

4. Metadata, Governance & Safety

License and format. CC-BY-4.0. Each trajectory is a self-contained JSON document following the 9-field schema (Appendix A), with per-step and aggregate metadata. A datasheet following Gebru et al. (2021) accompanies the release documenting collection methodology, intended uses, and known limitations.

Release tiers. (1) Full traces where provider terms of service permit redistribution of model outputs. (2) Abstracted state summaries (phase transitions, tool calls, error events, revision counts—without verbatim model reasoning) where full release is restricted. Abstracted summaries preserve $>90\%$ of analytical utility for error localization, process quality scoring, and autonomy classification, since these tasks depend on structured metadata rather than raw text.

Privacy: No PII; all tasks use public scientific knowledge. No dual-use content. **Splits:** Task-level 80/20 train/test, domain-stratified, no leakage (Kapoor & Narayanan, 2023). Evaluation follows HELM (Liang et al., 2023).

5. Acceleration Potential

OpenDiscoveryTrace enables six capabilities currently unavailable to the community. (1) *Process-aware agent evaluation*: benchmarking that distinguishes systematic reasoning from lucky guessing, complementing output-only metrics (Wang et al., 2023; Gil et al., 2014). (2) *Early failure intervention*: our finding that 68.1% of first errors occur at step 0 (the initial tool invocation) suggests that monitoring early trajectory steps could prevent the majority of downstream failures. (3) *Error-type-specific model improvement*: the discovery that Claude's errors are predominantly tool misuse while GPT-5.4's are reasoning errors implies different training strategies for each architecture. (4) *Process-aware reward modeling*: reward shaping over full trajectories rather than terminal outcomes, enabling RL approaches that credit sound methodology independent of final result. (5) *Trustworthy AI scientist auditing*: complete decision traces for governance of autonomous research systems, a prerequisite as AI agents move toward safety-critical scientific domains (King et al., 2009; Kitano, 2021). (6) *Reproducible open-weight benchmarking*: 120 trajectories from 4 open-weight models enable fully reproducible comparison without proprietary API access, bridging the evaluation gap between open and closed models.

OpenDiscoveryTrace shifts evaluation from *what* AI scientists produce to *how* they reason—a prerequisite for trustworthy autonomous science.

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Appendix

A. Complete Trace Schema

Every trajectory in OPENDISCOVERYTRACE is stored as a self-contained JSON file. The schema below defines every field. Fields marked $\star$ are present in every trajectory; others may be `null` for certain steps.

Listing 1. Complete OPENDISCOVERYTRACE JSON schema for a single trajectory.

```
{
  "trajectory_id": "string* (unique: {task_id}_{model})",
  "task_id": "string* (e.g. dd_e01, ms_m03, gn_h02, lt_m05)",
  "domain": "string* (drug_discovery | materials_science
                      | genomics | literature)",
  "difficulty": "string* (easy | medium | hard)",
  "prompt": "string* (the scientific task given to the model)",
  "ground_truth": "string (expected answer for easy tasks; null for
                      medium/hard open-ended tasks)",
  "model": "string* (gpt-5.4 | claude-opus-4.6 |
                      gemini-3.1-pro | qwen2.5-7b |
                      mistral-7b-v0.3 | phi-3.5-mini |
                      qwen2.5-1.5b)",
  "open_weight": "boolean (true for open-weight models)",

  "trajectory": [
    {
      "step_id": "integer* (0-indexed sequential step number)",
      "timestamp": "string* (ISO 8601 UTC, e.g. 2026-04-16T14:30:00Z)",
      "phase": "string* (literature_review | hypothesis |
                       experiment_design | execution |
                       analysis | revision | conclusion)",
      "thought": "string* (model's reasoning at this step)",
      "action": {
        "type": "string* (tool_call | reasoning | conclude)",
        "tool": "string (python_exec | web_search |
                      pubmed_search | api_call | none)",
        "input": "string (argument passed to the tool)",
        "output": "string (raw output returned by the tool)"
      },
      "observation": "string (processed observation from action)",
      "error": {
        "occurred": "boolean* (true if this step had an error)",
        "type": "string (tool_error | api_error | timeout |
                       parse_error | null)",
        "message": "string (error message text; null if no error)"
      },
      "revision_trigger": "string (what prompted a strategy change;
                          null if no revision at this step)",
      "confidence": "float (model's self-reported certainty
                          [0.0, 1.0]; null if not stated)",
      "raw_response": "string (first 3000 chars of model output)",
      "wall_time": "float (seconds for this step)"
    }
  ],

  "outcome": {
    "success": "boolean (true/false/null; null = not yet
                          evaluated against ground truth)",
    "final_claim": "string (model's concluding statement)",
    "confidence": "float (confidence on the final claim)",
    "verification": {
      "method": "string (ground_truth_match | llm_judge | pending)",
      "result": "string (correct | incorrect | partial | pending)",
      "score": "float (0.0--1.0 verification score)"
    },
    "failure_type": "string (from failure taxonomy; null if
                          no failure)",
    "recovery_attempted": "boolean (true if the model tried to recover
                          from an error)",
    "recovery_successful":"boolean (true if recovery led to success;
                          null if no recovery attempted)"
  },

  "metadata": {
    "total_steps": "integer* (number of steps in trajectory)",
    "total_tokens_est": "integer (estimated token count)",
    "total_tool_calls": "integer* (number of tool invocations)",
    "total_failures": "integer* (number of steps with errors)",
    "total_revisions": "integer* (number of strategy revisions)",
```

Table 1. Comparison of OPENDISCOVERYTRACE with existing AI science benchmarks. OPENDISCOVERYTRACE is the first benchmark to capture full process traces including error logging, revision tracking, and failure analysis alongside final output evaluation. ✓ = supported, × = not supported, ∼ = partial support.

Benchmark	Process Traces	Error Logging	Revision Tracking	Multi-Domain	Science Specific	Multi-Model
ScienceAgentBench (Chen et al., 2025)	×	×	×	✓	✓	✓
DiscoveryBench (Majumder et al., 2024)	×	×	×	✓	✓	✓
The AI Scientist (Lu et al., 2024)	×	×	×	∼	✓	✓
PaperBench (Starace et al., 2025)	×	×	×	∼	✓	✓
ResearchBench (Liu et al., 2025)	×	×	×	✓	✓	✓
SciCode (Tian et al., 2024)	×	×	×	✓	✓	✓
MLAgentBench (Huang et al., 2024)	∼	∼	×	×	∼	✓
MLE-bench (Chan et al., 2025)	×	×	×	×	×	✓
WebArena (Zhou et al., 2024)	✓	∼	×	×	×	✓
AgentBench (Liu et al., 2024)	✓	∼	×	×	×	✓
OPENDISCOVERYTRACE (ours)	✓	✓	✓	✓	✓	✓

```
    "wall_time_seconds": "float* (total wall-clock time)",
    "max_steps_reached": "boolean (true if hit 30-step limit)",
    "model_version": "string (exact model ID used)",
    "temperature": "float (sampling temperature; 0.0 for
                          frontier, varies for open-weight)",
    "collection_timestamp":"string (ISO 8601 collection date)"
  }
}
```

B. Benchmark Comparison with Existing Datasets

Table 1 compares OPENDISCOVERYTRACE with existing AI science benchmarks and agent trajectory datasets. The key distinction is that OPENDISCOVERYTRACE is the only resource providing structured process traces with error logging, revision tracking, and multi-phase scientific workflow annotation alongside final output evaluation. Most existing benchmarks evaluate final artifacts only; agent trajectory datasets like ToolBench and MINT record traces but exclusively for web or software domains, not science.

C. Dataset Summary Statistics

D. Inter-Model Comparison

The central finding is that **output metrics (success rate, conclusion rate) show no significant differences**, while **process metrics (errors, revisions, wall time) show highly significant differences** with large effect sizes. An output-only benchmark would rate these three models as interchangeable; our process traces reveal fundamentally different error-handling behaviors.

E. Failure Taxonomy

We categorize errors into three types based on automated classification of error messages and context:

- **Tool misuse**: Incorrect API call, wrong parameters, malformed input to a tool, or misinterpretation of tool output. Example: passing a gene name to PubChem (which expects a chemical compound name) or calling `python_exec` with syntactically invalid code.
- **Reasoning error**: Logical fallacy, incorrect inference, flawed calculation, or unsupported conclusion. Detected via keyword-based revision triggers ("mistake", "reconsider", "let me correct").
- **Hallucination**: Agent fabricates data, citations, or tool outputs. Rare in our data (1 instance across 688 total errors).

Table 2. Complete dataset statistics.

Metric	Value
Scale	
Total trajectories	522
Frontier model trajectories	372 (124 $\times$ 3 models)
Open-weight model trajectories	120 (30 $\times$ 4 models)
Live-retrieval variant trajectories	30
Models represented	7 (3 frontier + 4 open-weight)
Domains covered	4
Tasks executed per frontier model	124
Tasks in full bank (designed)	200
Tasks per domain per frontier model	31
Trajectory characteristics (frontier models)	
Conclusion rate	99.7%
Success rate (evaluable subset, n=127)	69.3%
Mean steps per trajectory	4.1
Median steps per trajectory	2.0
Mean tool calls per trajectory	3.0
Mean errors per trajectory	1.0
Mean revisions per trajectory	0.87
Mean wall time (seconds)	99.1
Trajectories with ≥ 1 error	31.2%
Trajectories with ≥ 1 revision	34.4%
Token and cost estimates	
Estimated total tokens (frontier)	$\sim$701K
Mean response chars per trajectory (Claude)	11,237
Mean response chars per trajectory (GPT-5.4)	5,720
Mean response chars per trajectory (Gemini)	5,663
API cost (frontier models)	$\sim$\$300
GPU cost (open-weight models, Modal A10G)	$\sim$\$40

Key insight: Models differ not only in *how many* errors they make but in *what kind*. Claude's errors are predominantly tool misuse — it attempts more complex tool interactions and fails more often at the tool interface. GPT-5.4 makes far fewer errors overall, and when it does err, the failures are reasoning-level rather than tool-level. This distinction has direct implications for model improvement: Claude would benefit most from better tool-use training, while GPT-5.4's rarer errors require reasoning-level interventions.

Limitation: Error classification uses automated heuristics (keyword matching on error messages and revision triggers). This may undercount implicit reasoning errors that do not trigger explicit revision keywords. Future work will validate against human-labeled error types on the annotated subset.

F. Matched-Task Within-Domain Comparisons

To control for task and domain effects, we compare models only on shared tasks within each domain (31 tasks per domain, all three frontier models execute every task).

The error rate divergence is **significant in every domain** ($p<0.0001$), while step counts are **non-significant in every domain**. This confirms that the error-handling difference is robust to domain effects and not an artifact of task selection or difficulty confounds.

G. Difficulty $\times$ Model Interaction

The error-handling divergence is concentrated in **medium and hard tasks**. Easy tasks produce uniformly short, low-error trajectories across all models (Claude averages just 0.27 errors on easy tasks vs. 4.14 on medium). This validates both the difficulty stratification and the intuition that process differences emerge most strongly on challenging scientific problems.

Table 3. Complete inter-model comparison for frontier models (n=124 per model). All tests are Kruskal-Wallis (equivalent to Mann-Whitney U for pairwise). Cliff's δ effect sizes: negligible (<0.147), small (0.147–0.33), medium (0.33–0.474), large (>0.474). Values: mean ± SD. *** p<0.001, ** p<0.01, * p<0.05, n.s. not significant.

Metric	GPT-5.4	Claude Opus 4.6	Gemini 3.1 Pro	Significance	Cliff's δ (max pair)
Total steps	3.5 ± 3.5	4.1 ± 4.0	4.8 ± 5.4	n.s. (p=0.158)	0.160 (small)
Tool calls	2.5 ± 3.5	2.8 ± 3.2	3.8 ± 5.2	n.s. (p=0.214)	0.149 (small)
Errors	0.08 ± 0.3	2.5 ± 3.4	0.4 ± 0.8	*** (H=122, $p<10^{-4}$)	0.613 (large)
Revisions	0.4 ± 0.8	1.2 ± 2.0	1.0 ± 2.5	*** (H=15.6, p=0.0004)	0.244 (small)
Wall time (s)	39.6 ± 37.4	190.6 ± 148.3	67.0 ± 89.9	*** ($p<10^{-4}$)	—
Response chars	5,720	11,237	5,663	—	—
Success rate	69.0%	69.8%	69.0%	n.s. (p=0.997)	—
Conclusion rate	100%	99.2%	100%	n.s. (p=0.368)	—

Table 4. Failure taxonomy breakdown by model.

Model	Total Errors	Tool Misuse	Reasoning Error
Claude Opus 4.6	462	308 (66.7%)	153 (33.1%)
Gemini 3.1 Pro	165	46 (27.9%)	119 (72.1%)
GPT-5.4	61	10 (16.4%)	51 (83.6%)
Total	**688**	**364 (52.9%)**	**323 (46.9%)**

H. Benchmark Task Baselines

H.1. Task 1: Trajectory Outcome Prediction

Setup: Given all step-level features from a complete trajectory, predict binary success (evaluable subset, n=127). Features: total steps, tool calls, errors, revisions, unique tools, unique phases, response length, error type count.

Finding: No method exceeds the majority baseline accuracy (0.693). AUROC values are near or below chance. This confirms that **outcome prediction from trajectory-level features is genuinely difficult**: neither aggregate statistics (LR, RF, GBT) nor sequential patterns (LSTM, Transformer) on the current 8-dimensional feature set are predictive of success. This motivates future work on richer representations — e.g., semantic embeddings of thought content or graph-structured encodings of tool-call dependencies.

H.2. Task 2: Error Localization

Setup: Among the 116 trajectories containing at least one error, identify the first step where reasoning went wrong.

Result: 68.1% of first errors occur at step 0 (the first tool invocation). The "always predict step 0" heuristic achieves 68.1% accuracy; "always predict last step" achieves 0%. Random baseline: ∼7% (1/mean-steps).

Caveat: The step-0 concentration partially reflects harness design (the first substantive action is typically a tool call). Future work should vary tool-initialization ordering to disambiguate harness artifact from generalizable agent behavior.

H.3. Task 3: Claim Verification

Success rates with Wilson score 95% confidence intervals on the evaluable subset:

- GPT-5.4: 69.0% [54.0%, 80.9%] (n=42)
- Claude Opus 4.6: 69.8% [54.9%, 81.4%] (n=43)
- Gemini 3.1 Pro: 69.0% [54.0%, 80.9%] (n=42)

The wide confidence intervals reflect the limited evaluable subset (only easy tasks have ground truth). Expanding ground-truth coverage to medium tasks is a Phase 2 priority.

Table 5. Matched-task within-domain comparisons (n=31 shared tasks per domain per model). All tests Kruskal-Wallis. *** $p<0.001$, n.s. not significant.

Domain	Metric	Claude Opus 4.6	Gemini 3.1 Pro	GPT-5.4	H	Sig.
Drug Discovery	Steps	4.1 ± 3.1	4.7 ± 5.6	4.2 ± 3.9	0.60	n.s.
	Errors	2.5 ± 2.8	0.4 ± 0.9	0.03 ± 0.2	29.5	***
Genomics	Steps	5.0 ± 5.8	5.9 ± 5.9	3.2 ± 3.4	2.70	n.s.
	Errors	3.3 ± 5.6	0.5 ± 1.0	0.0 ± 0.0	29.3	***
Literature	Steps	3.9 ± 4.0	4.3 ± 5.9	3.0 ± 3.5	3.80	n.s.
	Errors	2.0 ± 2.3	0.3 ± 1.0	0.1 ± 0.3	31.8	***
Materials Science	Steps	3.3 ± 1.8	4.5 ± 3.9	3.6 ± 3.5	0.44	n.s.
	Errors	2.1 ± 1.7	0.3 ± 0.5	0.2 ± 0.5	34.0	***

Table 6. Per-difficulty, per-model mean steps and errors.

Difficulty	Model	Mean Steps	Mean Errors
Easy (n=147)	Claude Opus 4.6	1.9	0.27
	Gemini 3.1 Pro	3.4	0.27
	GPT-5.4	2.0	0.00
Medium (n=153)	Claude Opus 4.6	5.8	4.14
	Gemini 3.1 Pro	5.0	0.45
	GPT-5.4	4.9	0.12
Hard (n=72)	Claude Opus 4.6	4.9	3.54
	Gemini 3.1 Pro	7.4	0.42
	GPT-5.4	3.4	0.17

H.4. Task 5: Process Quality Scoring

Composite score: (efficiency + tool_use + low_errors + conclusion_rate) / 4, each normalized [0,1].

- GPT-5.4: 0.747 ± 0.008
- Gemini 3.1 Pro: 0.747 ± 0.011
- Claude Opus 4.6: 0.724 ± 0.067

Kruskal-Wallis H=149.9, $p<0.0001$. Claude scores significantly lower due to its higher error rate, despite matching success rates. This demonstrates that **process quality scoring reveals quality differences invisible to binary success metrics**.

I. LLM-Based Inter-Annotator Agreement

We annotated a 60-trajectory stratified sample (5 per model per domain) using GPT-5.4-mini and Claude Sonnet 4.6 as annotators. Each annotator rated every trajectory on four axes (1–5 ordinal scale for the first three; L1–L4 categorical for autonomy).

The autonomy taxonomy (L1–L4) exceeds the $\alpha \geq 0.67$ target, providing initial evidence that the proposed levels are reliably distinguishable. Weighted agreement (within 1 point on 5-point scales) exceeds 0.85 across all axes. We note that LLM-based annotation is a proxy for human annotation; the planned Phase 3 human expert study (3 annotators, 120 trajectories) will provide definitive IAA measurements.

J. Tool Usage and Latency Statistics

Python execution dominates (78.5% of 1,116 total calls). GPT-5.4 uses a notably more diverse tool mix (23.9% web search, 11.6% PubMed, 10.3% API calls) compared to Claude (98% Python). Mean per-tool latencies: `python_exec`

Table 7. Outcome prediction baselines (5-fold stratified CV).

Method	Accuracy	AUROC
Majority baseline	0.693	0.500
Logistic Regression	0.685 ± 0.046	0.496 ± 0.096
Random Forest	0.614 ± 0.049	0.452 ± 0.091
Gradient Boosting	0.614 ± 0.033	0.426 ± 0.066
LSTM (2-layer, h=64)	0.693 ± 0.015	0.422 ± 0.124
Transformer (2-layer, 4-head)	0.693 ± 0.015	0.369 ± 0.093

Table 8. LLM-based inter-annotator agreement (Krippendorff's α).

Axis	α	Interpretation
Correctness	0.629	Moderate
Reasoning quality	0.711	Substantial
Tool use efficiency	0.717	Substantial
Autonomy level (L1–L4)	0.820	Near-perfect

27.5s, `pubmed_search` 11.2s, `web_search` 8.8s, `api_call` 6.8s.

Note on web search: The current release uses simulated web search (models use parametric knowledge). The 30 live-retrieval variant trajectories demonstrate the harness's ability to execute real PubMed and PubChem API calls; PubMed returned results for 2 of 30 pilot tasks. Phase 4 will expand live retrieval to all literature-domain tasks.

K. Open-Weight Model Details

To enable fully reproducible baselines without proprietary API access, we generated 120 trajectories from four open-weight models, each executing the same 30 tasks (first 30 from the task bank: drug discovery easy and medium).

All open-weight models produce single-response trajectories (the entire scientific workflow in one generation) since they lack the multi-turn tool-call scaffolding of the frontier model harness. These trajectories are included to demonstrate that the schema accommodates any model interface and to provide a fully reproducible reference set for the community. Future work will implement tool-call support for open-weight models via function-calling APIs.

L. Task Bank Examples

The full task bank comprises 200 tasks (50 per domain). Below are representative examples at each difficulty level. The complete task bank is available in `task_bank.json` in the release.

Drug Discovery.

- **Easy**: "Retrieve the molecular weight, LogP, and number of hydrogen bond donors for aspirin. Report the exact values."
- **Medium**: "Find all known kinase inhibitors approved by the FDA that target VEGFR-2. For each, report the drug name, approval year, and whether it also inhibits other kinases."
- **Hard**: "Propose a novel drug repurposing hypothesis for an existing FDA-approved drug to treat Alzheimer's disease. Base your hypothesis on molecular target analysis, pathway overlap, and available clinical evidence. Design a computational experiment to validate your hypothesis."

Materials Science.

- **Easy**: "What is the band gap of gallium arsenide (GaAs)? Is it a direct or indirect band gap semiconductor?"
- **Medium**: "Design a high-entropy alloy (HEA) composition for high-temperature applications. Select 5 elements, justify each choice based on Hume-Rothery rules, and predict the likely crystal structure using the VEC criterion."

Table 9. Tool invocation counts by model (across 124 frontier trajectories each).

Tool	Claude	Gemini	GPT-5.4
`python_exec`	341	372	163
`web_search`	7	49	72
`pubmed_search`	1	41	35
`api_call`	1	3	31
Total	350	465	301

Table 10. Open-weight model specifications and performance.

Model	Parameters	GPU	Mean time/task
Qwen2.5-7B-Instruct	7B	A10G	∼15s
Mistral-7B-Instruct-v0.3	7B	A10G	∼15s
Phi-3.5-mini-instruct	3.8B	A10G	<1s
Qwen2.5-1.5B-Instruct	1.5B	V100	∼7s

- **Hard**: "Propose a novel solid-state electrolyte composition for all-solid-state lithium batteries that balances ionic conductivity, electrochemical stability window, and mechanical properties. Justify using first-principles reasoning and literature evidence."

Genomics.

- **Easy**: "What is the chromosomal location and function of the TP53 gene? How many exons does it have?"
- **Medium**: "Design a CRISPR guide RNA to knock out the human PCSK9 gene. Specify the target exon, the 20-nt guide sequence, the PAM site, and predict off-target effects."
- **Hard**: "Propose a computational pipeline to identify novel synthetic lethal gene pairs in pancreatic cancer. Design the approach using DepMap, COSMIC, and STRING, specify the statistical framework, and predict at least one novel interaction."

Scientific Literature.

- **Easy**: "Find the original paper describing the transformer architecture ('Attention Is All You Need'). Report the full citation, venue, and citation count."
- **Medium**: "Identify contradictions in the literature about whether transformer-based protein language models can predict protein function more accurately than alignment-based methods. Find papers supporting each side."
- **Hard**: "Write a critical analysis of the claim that 'AI will replace human scientists within 10 years.' Gather evidence from proponents and skeptics, evaluate argument strength, and formulate an evidence-based position."

M. Agent Harness Details

The agent harness is a Python script (`agent_harness.py`) that orchestrates trajectory generation. Key design choices:

- **System prompt**: A standardized 6-phase scientific workflow (literature review → hypothesis → experiment design → execution → analysis → conclusion). All models receive the identical system prompt.
- **Tool suite**: Four tools: `python_exec` (sandboxed subprocess with 60s timeout), `web_search` (simulated), `pubmed_search` (real E-utilities API), `api_call` (PubChem, UniProt, Materials Project).
- **Step limit**: Maximum 30 steps per trajectory. Models are nudged to conclude starting at step 15 and explicitly prompted at step 25.

- **Temperature**: 0.0 for all frontier models (deterministic). Open-weight models use `do_sample=False`.
- **Checkpointing**: Each trajectory is saved as a JSON file immediately upon completion.
- **Concurrency**: Up to 4 concurrent API calls per model to respect rate limits.
- **Error handling**: API errors (timeouts, rate limits) are logged in the trajectory and the model is given an opportunity to recover.

Sensitivity to harness design: The 6-phase prompt, 30-step cap, and tool suite are design choices that affect trajectory characteristics. Models may collapse phases when the prompt encourages comprehensive responses (as observed with Claude's single-response pattern in our earlier pilot). Future work should systematically vary these parameters (step limits of 10/30/60, alternative prompt structures, expanded tool suites) to quantify sensitivity.

N. Reproducibility

Code, data, and the complete task bank will be publicly released upon acceptance. The release includes an agent harness script for trajectory generation, analysis pipelines for reproducing all figures and statistics, and the full set of 522 trajectory JSON files.

Reproducing frontier trajectories requires API keys for the three frontier model providers. The harness script accepts a `-model` flag to select any supported model and a `-max-tasks` flag to control the number of tasks.

Reproducing open-weight trajectories requires only a single GPU (V100 16GB or better) and access to open-weight model weights. No proprietary API keys are needed, enabling fully independent replication.

Running analysis: A single script regenerates all figures, statistical tables, and benchmark baselines from the raw trajectory JSON files, ensuring end-to-end reproducibility of all reported results.